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Interstellar Travel

Professor Brand tells Cooper, in their first meeting, that the Lazarus missions have been sent out to search for new homes for humanity. Cooper responds, “There’s no planet in our solar system that can support life, and it’d take a thousand years to reach the nearest star. That doesn’t even qualify as futile. Where did you send them, Professor?”

The worse-than-futile challenge, if you don’t have a wormhole, is obvious when you realize just how far it is to the nearest stars (Figure 13.1).

Distances to Nearest Stars

The nearest star (other than our Sun) thought to have a habitable planet is Tau Ceti, 11.9 light-years from Earth, so traveling at light speed you would need 11.9 years to reach it. If there are any habitable planets closer than that, they can’t be much closer.